

Designing Clinical AI as a Managed System: A Systems Design and Management Case Study from MIT

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May 28, 2026

Abstract

This paper uses a project from MIT’s System Design and Management program to examine how clinical AI should be designed as a managed system, not as a standalone model. The case, Sanguine, is a physician-facing interpretation layer for Mayo Clinic laboratory results. Across the project, the team moved from stakeholder analysis and problem framing to architecture decisions, tradespace exploration, uncertainty modeling, development planning, and risk mitigation. The main finding is practical: in clinical AI, model capability is only one part of the design problem. Evidence grounding, release control, workflow fit, regulatory readiness, and institutional trust can dominate the final architecture and the project baseline. The resulting recommendation favors a higher-cost, risk-mitigated plan because the lower-cost alternatives preserve risks that could block clinical validation or damage trust.

Keywords: systems design, systems engineering, engineering management, complex systems, MIT SDM

1 Introduction

Many clinical AI proposals start with the model. They ask which model is most accurate, which retrieval method should be used, or which interface can make the answer easier to read. Those questions matter, but they are too narrow. In a clinical setting, an AI system also has to fit into an institution, a regulatory environment, a workflow, a risk posture, and a set of professional norms. A technically impressive model can still fail if it creates liability, adds work for clinicians, produces explanations that cannot be audited, or reaches patients in the wrong way.

This paper develops that argument through a case study from MIT’s System Design and Management (SDM) program. SDM combines system architecture, systems engineering, and project management in an integrated core built around complex socio-technical problems [1]. The case is Sanguine, a proposed physician-facing clinical interpretation layer for Mayo Clinic laboratory results. The concept uses an agentic AI clinical librarian to augment physician-facing test results with relevant research, while keeping diagnosis and final clinical judgment with the physician.

The paper is not a claim that Sanguine should be deployed as described. It is a first synthesis of the SDM work into a research-style article. The value of the case is in the design path: how the team narrowed the system boundary, selected architecture decisions, compared alternatives, modeled uncertainty, chose a development approach, and revised the baseline after risk analysis.

The paper makes three contributions. First, it documents a clinical AI architecture case in which governance and release control are treated as first-order design decisions. Second, it shows how

tradespace exploration changes when residual risk is added alongside cost and schedule. Third, it extracts a set of lessons for designing AI systems in domains where trust, validation, and institutional accountability are not optional.

2 Background

MIT describes SDM as an integrated approach to architecting, engineering, and managing complex products and systems [1, 2]. The integrated core emphasizes system architecture, systems engineering, and project management. That mix is useful for AI in health care because the system boundary rarely stops at the algorithm. It includes data access, clinical workflow, evidence management, validation, release policy, operational monitoring, and institutional responsibility.

The Sanguine project sits in a specific class of systems: AI-enabled clinical decision support. These systems are not pure automation. They sit between information production and professional judgment. A laboratory result is already a clinical artifact, but its value depends on interpretation. The design question is therefore not simply, “Can AI explain a result?” The better question is, “What kind of interpretive layer can increase clinical value without weakening safety, accountability, or workflow fit?”

Several system concepts are useful here. System architecture connects function, form, and concept [3]. Tradespace exploration makes competing architectures visible by comparing performance, cost, risk, and constraints. Design structure matrices help identify coupled decisions and change-propagation paths. Lifecycle thinking forces the team to consider validation, operation, adaptation, and eventual replacement rather than only launch. Risk management keeps the design honest by asking what happens if assumptions fail.

The case also draws on a common SDM pattern: start broad enough to understand the problem, then narrow the boundary enough to build a defensible architecture. That narrowing is not a loss of ambition. It is a design decision.

3 Method

This paper uses a reflective single-case study based on the Sanguine opportunity sets, project charter, architecture assignments, uncertainty analysis, development planning work, project model, risk analysis, final presentation materials, and executive memo completed in the MIT SDM core [4–12]. The case was selected because it contains a complete sequence from problem framing to executive recommendation.

The analysis follows the project chronologically, but the paper reorganizes the material around six design questions:

1. What is the system boundary?
2. Which stakeholders define value and feasibility?
3. Which architecture decisions drive performance, cost, and governance?
4. How do uncertainty and risk change the preferred architecture?
5. What development approach fits the lifecycle risks?
6. What baseline should leadership authorize?

The source material includes numerical models and estimates. Those numbers should be read as planning estimates, not as audited financial or clinical results. Their purpose in this paper is to show how relative decisions changed under different assumptions.

4 Case study: Sanguine

The case is Sanguine, a physician-facing clinical interpretation layer for Mayo Clinic laboratory results developed during the MIT SDM integrated core. The project began with a broad opportunity: Mayo Clinic Laboratories produces high-quality tests at large scale, including for external patients and physicians, but the result-delivery layer often looks like a commodity report. A result may include a value, a flag, and a reference range. The clinical interpretation still depends on the physician’s ability to connect that result to patient history, prior results, current evidence, and specialist knowledge.

The working system problem statement became: improve Mayo Clinic laboratory patient care outcomes at scale by augmenting physician-facing test results with relevant research using an agentic AI clinical librarian [6, 7]. This boundary mattered. Earlier versions of the concept included patient-facing communication, result explanation, follow-up support, and broader care navigation. Those ideas were not wrong, but they pulled the project toward a general medical assistant. The narrower boundary made the design problem more tractable and reduced liability: Sanguine would support physicians, not replace them or diagnose patients directly.

4.1 Stakeholders and needs

The stakeholder map included patients, external physicians, Mayo Clinic Laboratories, Mayo physicians, Mayo leadership, legal and compliance teams, regulators, software developers, lab technicians, insurers, and researchers [4, 5]. The early analysis surfaced a point that remained stable across the project: legal and regulatory stakeholders were not merely late-stage reviewers. They shaped the feasible design space.

Patients needed clearer interpretation, reassurance, and better preparation for conversations with clinicians. External physicians needed specialty-informed material without extra workflow burden. Mayo Clinic leadership needed a path that could scale the Mayo Clinic Model of Care without damaging trust. Legal and regulatory stakeholders needed evidence of compliance, traceability, and accountability. These needs led to metrics such as clinical accuracy, physician time-to-insight, compliance evidence coverage, patient comprehension, and cost per interpreted result [6].

4.2 Architecture decisions

The architecture work translated stakeholder needs into a small set of decisions. Six decisions carried most of the architectural weight: model origin, research paradigm, number of AI roles per result, data sources, clinical validation and release control, and model specification [7]. Other decisions, such as communication channel and low-level integration mechanism, were treated as design details once their sensitivity and coupling appeared lower.

The preferred architecture used a Gemini-family model on Google Cloud, retrieval-augmented generation plus fine-tuning, a large group of specialized AI roles, multiple internal and external clinical sources, tiered release control, and a powerful multimodal model for complex reasoning. The concept was called “Maximalist” in the course work, but the name is a little misleading. It was not maximal in every direction. It was maximal where reliability, grounding, and clinical review mattered most, while avoiding the cost of full human review for every result.

The most important design choice was not the model family. It was the control structure around interpretation. A single chatbot-like assistant was cheap, but weakly grounded. A fully human-reviewed model matched current clinical assurance but did not scale. Sanguine sat between those poles: multiple specialist roles generated and reconciled an interpretation, evidence was traceable, and rare or complex cases escalated through tiered release control.

4.3 Tradespace exploration under uncertainty

The deterministic tradespace compared candidate architectures by estimated clinical accuracy and cost per interpreted result [7]. Low-cost concepts resembled current portal or rule-based result delivery. They were inexpensive but offered limited interpretive support. Human-in-the-loop concepts offered high assurance, but their cost scaled with physician review time. The preferred region combined AI-supported interpretation, evidence retrieval, role decomposition, and partial escalation.

The next step added uncertainty through Monte Carlo simulation [8]. Accuracy was modeled with beta distributions because it is bounded between zero and one. Cost was modeled with log-normal or triangular distributions because operational costs are positive and often right-skewed. The uncertainty analysis changed the reading of the tradespace. Heavy compliance concepts looked attractive in deterministic comparison, but became less attractive once escalation queues, review-load variability, and high-tier model usage were included. C1, the preferred “Maximalist” concept, and C6, a more customizable concept, remained robust candidates.

The analysis showed that clinical accuracy was most sensitive to AI role decomposition, data-source coverage, and release control. Cost per result was most sensitive to release control, model origin, and model specification [8]. This is a useful result because it tells a project team where not to pretend. The team could not resolve the whole architecture by choosing a better model. It had to learn how much evidence diversity helped, how specialist roles interacted, and what escalation rate the clinical workflow could tolerate.

4.4 Development model and baseline

The development plan combined stage gates, agile delivery, and spiral risk reduction [9, 10]. Stage gates kept the program aligned with clinical, regulatory, and launch milestones. Agile delivery supported software development across data, AI, cloud, interface, and quality teams. Spiral elements were used around high-uncertainty areas such as multi-agent interpretation, data harmonization, and release control.

The initial project model used teams, tasks, phases, products, productivity multipliers, and dependencies. Ten project variants were compared. Full outsourcing lowered labor cost but increased schedule duration because coordination overhead dominated. Overtime did not help once burnout was modeled. Workforce growth compressed schedule but at a steep cost. Strategic scaling, which added contractors only to non-critical development bottlenecks while keeping core functions internal, became the preferred baseline at about \$4.69M and 43 weeks [10].

That result did not survive the final risk pass unchanged. Risk modeling identified seven major risks: EHR/lab integration failure, AI clinical accuracy failure, IRB or regulatory delay, patient data compliance failure, clinical validation bottleneck, late interface rework, and team knowledge loss [11]. The first risk model was too elaborate, with arbitrary risk functions and cascading tasks. The team simplified it to effect and probability. That simpler model was easier to explain and use.

The full mitigation plan added early interface testing, data integration feasibility work, clinical validation readiness, regulatory pre-submission, and PHI compliance review. Residual risk fell from

179 to about 75, a 58% reduction, but the plan moved from roughly \$4.8M and 46 weeks to \$7.1M and 50 weeks [11]. The decision was not Pareto-optimal in cost and schedule alone. It became defensible only when residual risk was treated as a third planning dimension.

4.5 Final recommendation

The executive recommendation was to authorize a \$7.1M, 50-week implementation program, with a prototype in Q4 2026, a private pilot in Q1 2027, and a controlled US launch for three test types in Q3 2027 [12]. The recommendation required three institutional commitments: Mayo IT capacity for controlled access to clinical and laboratory systems, regulatory and IRB preparation capacity, and physician reviewer availability for shadow-mode validation.

The case ends with a managerial lesson rather than a technical punchline. For clinical AI, the cheapest plan that appears feasible may depend on a long chain of risks not materializing. Once patient safety, clinical trust, regulatory approval, and institutional reputation are included, front-loaded mitigation is not overhead. It is part of the system design.

5 Discussion

The Sanguine case is useful because it resists the usual AI shortcut. The answer is not “use a better model.” The model matters, but the system is shaped just as much by evidence access, validation policy, physician workflow, regulatory timing, and the cost of being wrong.

5.1 Boundary choice was a risk-control mechanism

The project began with a broad ambition to improve patient understanding of lab results. Over time, the boundary moved toward a physician-facing clinical librarian. That change reduced the apparent surface area of the product, but it made the architecture stronger. A patient-facing AI explanation system would have carried a different liability profile, stronger communication requirements, and a higher chance of being interpreted as medical advice. The physician-facing boundary preserved the core value of better interpretation while keeping final judgment inside the clinical workflow.

5.2 Governance was architectural

In many software projects, governance is treated as process around the system. In this case, governance became part of the architecture. Tiered release control, audit trails, shadow-mode validation, PHI handling, and escalation rules were not compliance afterthoughts. They determined whether the system could be trusted and whether it could scale.

This is one of the clearer SDM lessons from the case. Architecture is not only about components. It is about the allocation of responsibility across humans, machines, institutions, and rules.

5.3 Uncertainty changed the preferred design

The deterministic tradespace made high-assurance concepts look attractive. The uncertainty model complicated that view. Concepts with more human review had higher assurance, but they also had more cost variance and operational sensitivity. Escalation queues are not a detail in a clinical AI system. They are part of the product’s ability to operate.

The result was not a rejection of human review. It was a more precise use of it. Tiered release control became preferable because it treated review capacity as scarce and directed it toward rare, ambiguous, or high-risk cases.

5.4 Risk mitigation changed the baseline

The project model initially favored strategic scaling: targeted contractors for bottleneck development tasks, internal ownership of core clinical and integration work, and a 43-week baseline under the \$6M budget target. After risk analysis, that plan was no longer enough. The full mitigation baseline cost more and used the available schedule margin, but it reduced residual risk by more than half.

That shift is easy to criticize if cost and schedule are the only axes. It is much harder to criticize once patient safety and institutional trust are included. The lower-cost plan implicitly required several bad events not to happen. The higher-cost plan bought down the risks most likely to block clinical validation: integration failure, hallucination or accuracy failure, regulatory delay, PHI compliance gaps, and late interface rework.

5.5 Limits of the case

This article is based on project work and planning models, not on a deployed clinical system. The cost estimates, accuracy assumptions, and escalation rates require empirical validation. The case also depends on institution-specific conditions, including Mayo Clinic’s data environment, brand position, clinical review capacity, and cloud strategy. A different institution would likely make different choices.

Even with those limits, the case offers a general lesson. Clinical AI design should begin with the managed system, not the model. The question is not only what the model can say. The question is who can rely on it, under what evidence, with what controls, at what cost, and with what consequences if it is wrong.

6 Conclusion

The Sanguine case shows how an SDM approach changes the design of clinical AI. The project did not stop at model selection. It linked stakeholder needs, architecture decisions, uncertainty, development planning, and risk mitigation into one managed system.

The main conclusion is straightforward: for clinical AI, the safest architecture may not be the cheapest or fastest one, and the most capable model may not be the dominant design choice. The stronger baseline is the one that makes clinical accuracy, traceability, regulatory readiness, workflow fit, and institutional trust explicit.

The next version of this paper should add figures from the tradespace and DSM work, tighten the citations, and decide whether the article is positioned for systems engineering, engineering management, health AI governance, or an arXiv preprint across those categories.

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